

SOFTWARE REQUIREMENTS SPECIFICATION

Healthcare Clinic Booking Platform

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Authoritative reference for Software Quality Assurance Engineers, Frontend Developers, and Product Managers.

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Document Overview

1	Introduction	Purpose, scope, definitions
2	Overall Description	Product perspective and system workflow
3	Functional Requirements	Authentication, discovery, booking, payment, lifecycle
4	Non-Functional Requirements	Performance, usability, security, reliability
5	Requirement Traceability Matrix	Requirement-to-QA technique mapping

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) details the functional, structural, and behavioral requirements for the Healthcare Clinic Booking Platform (v1.0.0). It serves as an authoritative reference for Software Quality Assurance (SQA) Engineers, Frontend Developers, and Product Managers.

1.2 Scope

- User Authentication & Profile Registration.
- Doctor Discovery, Filtering, and Schedule Selection.
- Checkout, Promo Code Validation, and Payment Execution.
- Post-Booking Lifecycle Management and Cancellation.

1.3 Definitions & Acronyms

Term	Definition
SRS	Software Requirements Specification
BVA	Boundary Value Analysis
EP	Equivalence Partitioning
RBAC	Role-Based Access Control
SP	Story Points (Fibonacci Scale)

2. Overall Description

2.1 Product Perspective

The platform is a single-page web application using client-side storage (localStorage) for state management across user sessions.

2.2 System Workflow

Sign Up / Login	→	Search & Filter Doctors	→	Select Slot

My Bookings Dashboard	←	Confirmation	←	Checkout & Promo Code
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3. Functional Requirements

MODULE 1: AUTHENTICATION & ACCESS CONTROL

REQ-FUN-01	Patient Registration	5 SP
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User Story: As a new patient, I want to create an account using my personal details so that I can log in and book appointments.

Functional Specifications:

- Input Fields: Full Name, Phone Number, Email Address, Date of Birth, Gender, Password, Confirm Password.
- Full Name: Alphabetic characters and spaces only. Min 3, Max 50 chars.
- Email Address: Standard RFC 5322 format (user@domain.com). Must be unique.
- Phone Number: Exactly 11 digits starting with Egyptian carrier prefixes (010, 011, 012, 015). Must be unique.
- Date of Birth: Past date only. Age must be ≥ 18 years relative to system date.
- Password: Min 8, Max 20 chars; ≥ 1 uppercase, ≥ 1 lowercase, ≥ 1 digit, and ≥ 1 special character (!@#\$%^&*).
- Confirm Password: Must match Password field exactly.

Validation / Exception Handling:

- Invalid inputs trigger immediate inline errors and block form submission.

REQ-FUN-02	Patient Authentication	3 SP
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User Story: As a registered patient, I want to log in using my credentials so that I can access my profile and book appointments.

Functional Specifications:

- Input Fields: Email Address or Phone Number, Password. Both are mandatory.
- Invalid Credentials: Display generic message: "Invalid email/phone or password."
- Brute-Force Protection: 3 consecutive failed login attempts lock the account for 15 minutes.

Validation / Exception Handling:

- When locked, display: "Account locked due to multiple failed attempts. Please try again in 15 minutes."

MODULE 2: DOCTOR SEARCH & DISCOVERY

REQ-FUN-03	Doctor Search & Filtering	5 SP
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User Story: As a logged-in patient, I want to search and filter doctors by specialty, area, and price range so that I can find a suitable doctor.

Functional Specifications:

- Search Bar: Text input up to 100 characters. Dynamically filters by Doctor Name or Specialty.
- Specialty: Cardiology, Dermatology, Pediatrics, Dentistry.
- City/Area: Cairo - Nasr City, Cairo - Maadi, Giza - Dokki.

- Price Range: Numeric min/max inputs. Allowed range: 50 EGP to 2000 EGP.
- Price Validation: Reject queries where Min Price > Max Price.

Validation / Exception Handling:

- If no results match, display: "No doctors found matching your criteria."

REQ-FUN-04**Doctor Profile & Schedule Grid Selection****3 SP**

User Story: As a logged-in patient, I want to view a doctor's detailed profile and select an available time slot so that I can initiate my booking.

Functional Specifications:

- Profile Info: Doctor Name, Photo, Specialty, Consultation Fee, Address, Rating (1.0 to 5.0 stars).
- Schedule Grid: Displays available slots for the upcoming 7 calendar days only.
- Grouping: Slots are grouped by Date and Time (e.g., Monday, Oct 12 - 05:00 PM).
- Booked Slots: Previously booked slots are greyed out and marked as "Booked".

Validation / Exception Handling:

- Past dates and past time slots must be disabled.

MODULE 3: BOOKING ENGINE & CHECKOUT PROCESS

REQ-FUN-05**Patient Details & Promo Code Engine****5 SP**

User Story: As a logged-in patient, I want to enter patient details and apply promo codes to calculate the final consultation price.

Functional Specifications:

- Patient Selection: "Booking for Myself" (Default) vs "Booking for Someone Else".
- Beneficiary: Selecting "Booking for Someone Else" requires Beneficiary Name (Alpha, 3-50 chars) and Beneficiary Phone (11-digit EG format).
- Promo Input: Up to 15 chars; Alphanumeric + Hyphens. Input triggers trim and uppercase normalization.
- SAVE10: Deducts 10% from total fee.
- FLAT50: Deducts 50 EGP flat rate.
- BEST-QA: Deducts 100% (Total = 0 EGP).

Validation / Exception Handling:

- BEST-QA message: "Discount Applied! You are officially the best QA team. 🎉"
- Unrecognized codes trigger: "Invalid or expired promo code."
- Promo code cannot be applied more than once per booking session.
- Total fee after discount cannot fall below 0 EGP.

REQ-FUN-06**Payment Execution & Booking Finalization****5 SP**

User Story: As a logged-in patient, I want to select a payment method and finalize my booking to receive a confirmed booking receipt.

Functional Specifications:

- Payment Options: “Pay at Clinic (Cash)” or “Credit Card”.
- Card Number: Exactly 16 numeric digits.
- Expiry Date: MM/YY format and must be $\geq$ current month/year.
- CVV: Exactly 3 numeric digits.
- Transaction Finalization: Mutates slot state from Available $\rightarrow$ Booked.
- Reference ID: Generates a unique 8-character reference ID (e.g., #BK-89A12).
- Receipt: ID, Doctor, Date/Time, Address, Total Paid, and Method.

MODULE 4: POST-BOOKING LIFECYCLE MANAGEMENT

REQ-FUN-07	Appointment Dashboard & Cancellation	3 SP
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User Story: As a logged-in patient, I want to view my upcoming bookings and have the option to cancel them according to the cancellation policy.

Functional Specifications:

- Dashboard Views: “Upcoming” and “Past / Cancelled”.
- Cancellation Rule: “Cancel Booking” is active only if appointment time is > 24 hours away.
- State Mutation: Confirmation modal updates booking status to “Cancelled”.
- Slot Reversion: Slot automatically reverts from Booked $\rightarrow$ Available.

Validation / Exception Handling:

- If appointment time is ≤ 24 hours away, the button is disabled with tooltip: “Cancellations are only allowed up to 24 hours before the appointment.”

4. Non-Functional Requirements

NFR-01	Performance	Client-side interactions & filtering must render in < 200ms.
NFR-02	Usability	Responsive across Mobile (< 480px), Tablet, and Desktop.
NFR-03	Security	Sensitive data must not be exposed in raw text logs.
NFR-04	Reliability	LocalStorage state must persist reliably across reloads.

Quality Gates

- Functional acceptance should verify every REQ-FUN item against its stated validation and state-mutation rules.
- Responsive behavior should be verified at mobile, tablet, and desktop breakpoints.
- Performance acceptance should include measured client-side interaction and filtering latency.
- Security testing should confirm that sensitive credentials and payment data are not emitted in raw text logs.

5. Requirement Traceability Matrix (RTM)

REQ-FUN-01	Registration	Boundary Value Analysis, Equivalence Partitioning
REQ-FUN-02	Auth	State Transition Testing, Security Testing
REQ-FUN-03	Discovery	Equivalence Partitioning, Error Guessing
REQ-FUN-04	Discovery	State Transition Testing
REQ-FUN-05	Checkout	Decision Table Testing, Boundary Value Analysis
REQ-FUN-06	Payment	Equivalence Partitioning, Boundary Value Analysis
REQ-FUN-07	Management	State Transition Testing, Business Rule Testing

Traceability Notes

The RTM maps each functional requirement to the principal QA test-design or testing technique identified in the SRS. The mapping is intended to support test planning, coverage analysis, and defect investigation.

End of Document